Fully Covered Adversarial Camouflage Against Remote Sensing Detection via Physics-Driven Rendering and Pyramid Training

Ziyang Liu, Hongyuan Wang, Qianhao Ning, Zijian Wang, Yinxi Lu, Yunzhao Zang, and Zhiqiang Yan, *Member, IEEE*

***Abstract*—Uncrewed aerial vehicles (UAVs) imaging object detection systems based on deep neural networks (DNNs) are highly vulnerable to adversarial attacks. However, existing methods neither fully considered the complex physical interferences in the real world nor the multiscale and temporal observation characteristics of UAV-based detection, which makes the generated camouflage difficult to maintain effectiveness in practical tasks. To address these limitations, a fully covered robust adversarial camouflage generation framework for UAV-based detection models is introduced. First, the physically based differentiable renderer (PBDR) is designed to simulate global illumination, surface reflectance, and atmospheric scattering, thereby incorporating complex physical interference factors during camouflage training. Second, the pyramid camouflage training framework (PCTF) is introduced, which conducts hierarchical training on distance-varying inputs to preserve the multiscale adaptability of camouflage features across layers and ensure interframe consistency within layers. Furthermore, the limitations of existing evaluation indicators for UAV-based adversarial attacks are analyzed, and a novel set of indicators is proposed for evaluating physical adversarial camouflage in remote sensing. Extensive cross-domain experiments in both digital space and physical world demonstrate superior performance compared with state-of-the-art (SOTA) methods. The proposed approach achieves a temporal attack success rate (TASR) of 84.5%, see demo video at: https://youtube.com/@meyond_paa?si=foP1y4MNcC4ByZoi, compared with baselines, the average precision (AP) scores across multiple black-box detection models decrease by 18.4% on average, while the attack success rate (ASR) increases by 70.4%.**

***Index Terms*—Differentiable renderer, physical adversarial attack, pyramid training, remote sensing detection, uncrewed aerial vehicle (UAV).**

I. Introduction

With the development of uncrewed aerial vehicle (UAV)-based object detection models driven by deep neural networks (DNN) in the field of remote sensing [1], [2], adversarial attacks have simultaneously evolved due to their ability to expose model vulnerabilities and aid in deriving decision boundaries. These attacks have evolved from the early stage of digital adversarial attacks that directly added tiny perturbations to input images, to physical adversarial attacks that mislead detection models in the real world through physical forms, such as adversarial patches [3], [4] or fully covered adversarial camouflage attached to the target surface [5], [6], [7]. However, existing physical adversarial attacks for remote sensing object detection are mostly limited to near-distance scenarios and assume stationary targets, which makes the generated adversarial camouflage difficult to maintain effectiveness and robustness in practical remote sensing detection tasks [8], [9].

Building upon the background above, existing adversarial attack research on UAV-based detection models can be broadly categorized into digital adversarial attacks and physical adversarial attacks. Digital adversarial attacks typically achieve the attack effect by directly superimposing tiny perturbations onto the input images of detection models [10], [11]. However, in real-world applications, UAVs performing detection tasks are often noncooperative units, which makes it impractical to manipulate the captured images directly. Consequently, such digital adversarial attack methods cannot be realistically deployed in UAV-based detection tasks. In contrast, physical adversarial attacks generally employ physical forms such as fully covered adversarial camouflage attached to the target surface, thereby altering the input features perceived by detection models. Compared with digital adversarial attacks, physical adversarial attacks are of greater practical significance. Thus, this article primarily investigates physical adversarial attack methods against UAV-based object detection models.

In the study of physical adversarial attack methods for UAV-based detection models, the generated adversarial camouflage can be categorized according to its form into patch-based and fully covered types, as illustrated in Fig. 1(a). Most patch-based adversarial camouflage generation methods are still similar to digital adversarial attacks: perturbations are typically added to the target region in the images, which are then printed and directly attached to flat surfaces of the target. Some methods have made preliminary attempts to account for the differences between the digital space and the physical world, such as projection transformation [12] and illumination variation [8], [9]. These adversarial camouflage generation methods rely on input images, and their generalization across multiple views is limited to the shooting angles included in

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The authors are with the Research Center for Space Optical Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: meyondlzy@stu.hit.edu.cn; yanzhiqiang@hit.edu.cn).

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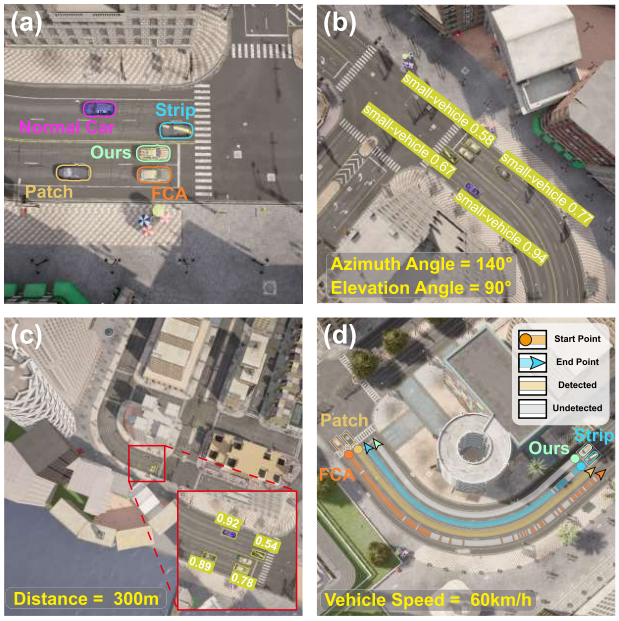


Fig. 1. Performance of adversarial camouflages generated by patch [13], strip [14], FCA [5], and ours. (a) Camouflaged targets with the corresponding generation methods labeled beside them. (b) Detection results under varying viewing angles. (c) Detection results under varying distances. (d) Detection results in dynamic scenarios. Bright regions indicate the detected areas, while light regions indicate the undetected areas.

the input images. Moreover, using patches as the camouflage medium has apparent limitations, as they cannot fully cover the target, leaving blind spots in the camouflage. Consequently, the performance of these camouflages deteriorates in real-world detection tasks with varying viewpoints [13], [14], as illustrated in Fig. 1(b). In contrast, fully covered adversarial camouflage does not suffer from these issues and can maintain high adversarial effectiveness across different viewing angles. Full-coverage camouflage is generated based on differentiable rendering [5], [15], [16]: starting from a 3-D physical model of the target, the target surface texture is optimized under constraints imposed by loss functions across different viewpoints and lighting conditions, ultimately producing a camouflage that fully covers the target.

Apart from the differences in camouflage forms, existing studies on physical adversarial attacks for remote sensing detection tasks continue to adopt training frameworks originally designed for near-distance scenarios [13]. They lack systematic modeling of feature shifts caused by changes in viewing angles and illumination, and feature degradation caused by increased distance and atmospheric scattering. When camouflage generated based on these methods is applied in real-world scenarios, the attack success rate (ASR) drops sharply, as shown in Fig. 1(c). Methods targeting dynamic scenarios remain limited to digital-level geometric augmentations [6], [8], such as rotation and random scaling of the original images, and fail to incorporate real physical variables arising from high-speed relative motion between UAVs and targets, including motion blur and instantaneous viewpoint shifts. Consequently, the generated camouflage cannot maintain continuous and stable deception in real UAV detection tasks, as shown in Fig. 1(d). More importantly, existing works generally use ASR as the only evaluation indicator [17], [18], [19], overlooking the temporal continuity in object detection tasks. In continuous detection scenarios, a correct detection in any frame of the sequence should be regarded as a successful detection. Indicators like ASR, which directly compute the ratio of successfully attacked frames to the total number of frames [20], [21], [22], [23], cannot adequately assess the temporal stability of the adversarial camouflage, further widening the gap between laboratory results and real-world deployment.

To address these issues, this article proposes a physical adversarial camouflage generation method capable of overcoming feature shifts caused by variations in viewpoint, scale, and illumination, and feature degradation in long-distance and dynamic scenarios, thereby enabling the transfer of physical adversarial camouflage from laboratory-level conditions to the complex and dynamic scenarios of UAVs. The specific contributions of this article are as follows.

1) A physically based differentiable renderer (PBDR) is developed to incorporate various real-world physical disturbances into camouflage generation through inverse rendering, thereby bridging the gap between the digital space and the physical world, enhancing the robustness of adversarial camouflage under complex illumination, varying viewpoints, and dynamic scenarios in real-world settings.
2) A pyramid camouflage training framework (PCTF) is designed to construct interlayer multiscale and intralayer multiviewpoint relationships in hierarchical training of camouflage generation, which enhances the adaptability of camouflage to scale and viewpoint variations during the training process and extends the maximum effective distance of adversarial camouflage to meet the requirements of real UAV-based detection tasks.
3) New evaluation indicators for real-world tasks are proposed to refine camouflage evaluation tailored to UAV-based object detection scenarios, systematically assessing the adversarial effectiveness of the camouflage. Extensive experiments across digital and physical domains demonstrate that the proposed camouflage generation method can successfully deceive object detection models in real-world tasks, achieving performance superior to state-of-the-art (SOTA) methods.

II. Related Works

This section first introduces physical adversarial attacks in remote sensing, followed by research progress on physical adversarial attacks in long-distance and dynamic scenarios.

A. Physical Adversarial Attack in Remote Sensing

In the field of remote sensing, physical adversarial attack methods can be categorized into patch-based and fully covered types. The patch-based adversarial camouflage was initially inspired by digital adversarial attack approaches [15], [24], [25], [26], where noise is added to the target region in the input image and then printed out and attached either onto

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flat surfaces of the target [27], [28], [29] or on the surrounding ground [30], [31]. Subsequently, researchers argued that this strategy adopts an oversimplified modeling during the digital-to-physical transfer, neglecting the imaging viewpoint information and the target's 3-D characteristics. To address this, Cui et al. [12] leveraged a projection transformation model to reduce texture feature shifts caused by varying camera perspectives, while Zhang et al. [8], [28] employed a simple differentiable model combined with the target's 3-D geometry, enabling the texture to adapt to subtle curvatures on relatively flat target surfaces during training. Nevertheless, patch-based adversarial camouflage remains inherently constrained by its design, making it difficult to cope with the angle variations in remote sensing tasks, such as UAV-based object detection.

Compared with the patch-based type, fully covered adversarial camouflage does not suffer these drawbacks. Its design, which covers the entire surface of the target, enables it to withstand detection from various viewpoints better. However, fully covered camouflage requires more detailed consideration of the physical properties of both the target and scene during training. Zhang et al. [32] employed a neural approximation function to simulate the effect of applying camouflage to the target surface. Wang et al. [5] utilized a neural mesh renderer (NMR) [15], along with a simple lighting model, to simulate changes in viewpoint and direct illumination during training. Meanwhile, Suryanto et al. [6], [33] applied a neural transformation network to extract scene lighting information from the target surface, and after perturbing the texture, reapplied this extracted information. This approach helps bridge the digital-to-physical gap, but still lacks rigorous physical modeling.

B. Adversarial Attack in Long-Distance Scenarios

Most existing adversarial attack methods are designed for near-distance scenarios. Du et al. [9] and Zhang et al. [8], [28] placed adversarial patches on the top and around the target to mislead UAV-based object detection models. Similarly, Lian et al. [30] and Wang et al. [31] attached patches only around the target, which induce the misclassification of camouflaged objects and generated false targets. Although these methods explored the applicability of adversarial camouflage in UAV-based detection tasks, they were mostly limited to near-distance scenarios (‘!100 m) and did not account for the impact of viewing distance in real tasks.

In adversarial attack research targeting UAV-based object detection in long-distance scenarios, Zhang et al. [13] achieved an ASR of up to 50% at 25–50-m distances. However, the ASR dropped sharply beyond 100-m distances, eventually losing adversarial effectiveness. Sun et al. [14] proposed a stripe-shaped camouflage that successfully attacked object detection models from certain viewing angles at distances up to 300 m, enhancing the adversarial performance in long-distance scenarios. Nevertheless, these methods either employ particular camouflage patterns, such as stripes, which are less prone to degradation at longer distances, or directly adopt camouflage training frameworks originally designed for near-distance scenarios. They failed to account for physical interference factors (e.g., atmospheric scattering) and sampling variations of camouflage across different distances [34], [35], which inevitably affect adversarial attacks in UAV-based scenarios and impose technical requirements on the mechanisms that preserve adversarial camouflage features across different shooting distances. Therefore, the lack of mechanism-level investigation has significantly undermined their adversarial effectiveness in real-world applications.

C. Adversarial Attack in Dynamic Scenarios

Currently, research on dynamic scenarios has mainly focused on adversarial attacks against human action recognition models. In the study of adversarial attacks targeting UAV-based object detection tasks, most works still concentrate on static targets, with some camouflage designs even placed on the ground surrounding the target. Although these methods can achieve relatively high adversarial effectiveness, their design inherently limits the application scenarios of camouflage.

Among the few studies that consider the motion state of targets, such as Du et al. [9] and Zhang et al. [28], the performance of camouflaged targets was evaluated during experiments under motion, without any specific design tailored for dynamic scenarios. Their results showed that although the overall ASR of the camouflage remained relatively high across entire video sequences in motion scenarios, the target was still correctly detected once every several frames or a dozen frames. As noted earlier, such occurrences would still be regarded as successful detections in real-world missions. This limitation arises from the lack of explicit modeling of motion characteristics, leaving these approaches unable to sustain adversarial performance under complex real-world interferences such as dynamic viewpoint shifts and motion blur.

In summary, existing physical adversarial attack methods in remote sensing rely solely on digital-level data augmentation for robustness enhancement. They lack explicit modeling of physical factors, such as complex illumination conditions, material reflectance properties, atmospheric scattering, and motion characteristics, and no adversarial camouflage generation framework has been established to address multiscale and multiview detection scenarios. As a result, camouflage rapidly fails under complex physical interferences in real tasks. Therefore, based on PBDR and PCTF, this article proposes a physically robust adversarial camouflage generation method for UAV-based object detection models, enabling the transfer of camouflage from laboratory-level experiments to the complex and dynamic conditions of real UAV operational scenarios.

III. Methodology

This section presents the generation method for fully covered physical adversarial camouflage targeting UAV-based object detection models. The methodological framework is illustrated in Fig. 2. Section III-A defines the problem and provides the background of adversarial camouflage generation. Section III-B introduces the PBDR model, which links the physical imaging process with differentiable rendering, offering an inverse rendering platform for generating robust

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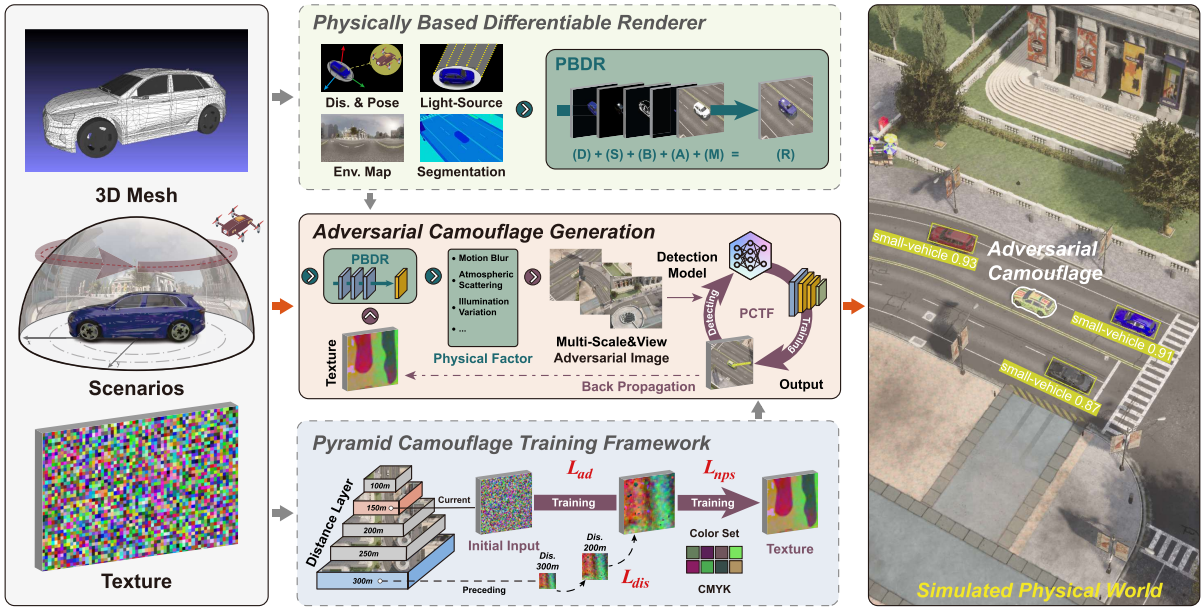


Fig. 2. Framework of this article.

physical adversarial camouflage. Section III-C describes the PCTF framework, which preserves texture features under multiview and multiscale transformations during training and, together with the designed loss function, produces the final adversarial camouflage.

A. Problem Definition

This article acquires customized simulated input images I from the CARLA simulator to meet the training requirements for input images under different viewing angles, distances, and lighting conditions. These images are captured from multiple positions with varying viewing angles θ_v, distances θ_d, and lighting conditions θ_l. The angle information θ_v includes the pose angles of both camera and target, and the lighting condition θ_l comprises the direct illumination from the light source as well as the indirect illumination from the global environment. Since the target to be camouflaged is unique throughout a single training process, all images share the same model mesh $\mathbf{M}$.

Therefore, in the kth texture optimization of an epoch, the input information is $(I^k, \theta_v^k, \theta_d^k, \theta_l^k, m, M, T^k)$. Feeding these inputs into the differentiable renderer $\mathbf{R}$ yields the rendered result of the camouflaged target during training. It should be noted that, to preserve texture features at different scales, data with varying distances θ_d^k are trained in separate batches, which will be elaborated in Section III-C. Meanwhile, to save computational resources, during the texture training process, the renderer only renders the texture-attached region of the target, while the other texture-free regions (e.g., glass and tires) and the background are extracted from the input image using the mask m, as shown in the following equation:

$$I_{\text{adv}}^k = m \cdot \mathbf{R}\left(M, T^k, \theta_v, \theta_d, \theta_l\right) + (1 - m) \cdot I^k. \tag{1}$$

In this article, the camouflage training is conducted under a white-box setting. During each camouflage texture optimization, the adversarial image I_{adv}^k is fed into the object detection model $\mathbf{D}$ to obtain the detection result $\mathbf{O}^k = \{\text{box}, \text{cls}, \text{conf}\}$, where box denotes the predicted location of the rotated anchor box (typically the coordinates of its four vertices), while cls and conf represent the predicted class label and confidence score of the target within that anchor box. Based on $\mathbf{O}^k$ and the current texture T^k, the loss function $\mathcal{L}$ can be computed, the detailed components of which will be described in Section III-C. Then, the backpropagation is applied to optimize the texture and generate the next input texture T^{k+1}. The proposed algorithm and process for attacking the object detection model can be simplified as follows:

$$T^{k+1} = \arg\min_{T^k} \left(\mathcal{L}\left(\mathbf{O}^k, T^k, G\right)\right). \tag{2}$$

Here, G is the ground truth of the input images.

B. Physically Based Differentiable Renderer

During adversarial camouflage training, the differentiable renderer reconstructs the optimized 2-D texture into 3-D space through forward rendering. It models light propagation, scattering, and reflection using a physics-based rendering pipeline, while incorporating motion characteristics and an atmospheric scattering model to simulate real-world illumination conditions and imaging degradation effects. Subsequently, the inverse rendering is performed to back-propagate gradients based on the detector outputs for further texture optimization.

However, other methods typically employ simplified, differentiable renderers during training, causing the composite camouflaged image computed via (2) to oversimplify the illumination, reflection, and imaging differences between the camouflaged target and its background. This simplification prevents accurate modeling of radiative transfer and visual degradation in the physical imaging process, and thus, real-world physical perturbations cannot be effectively propagated

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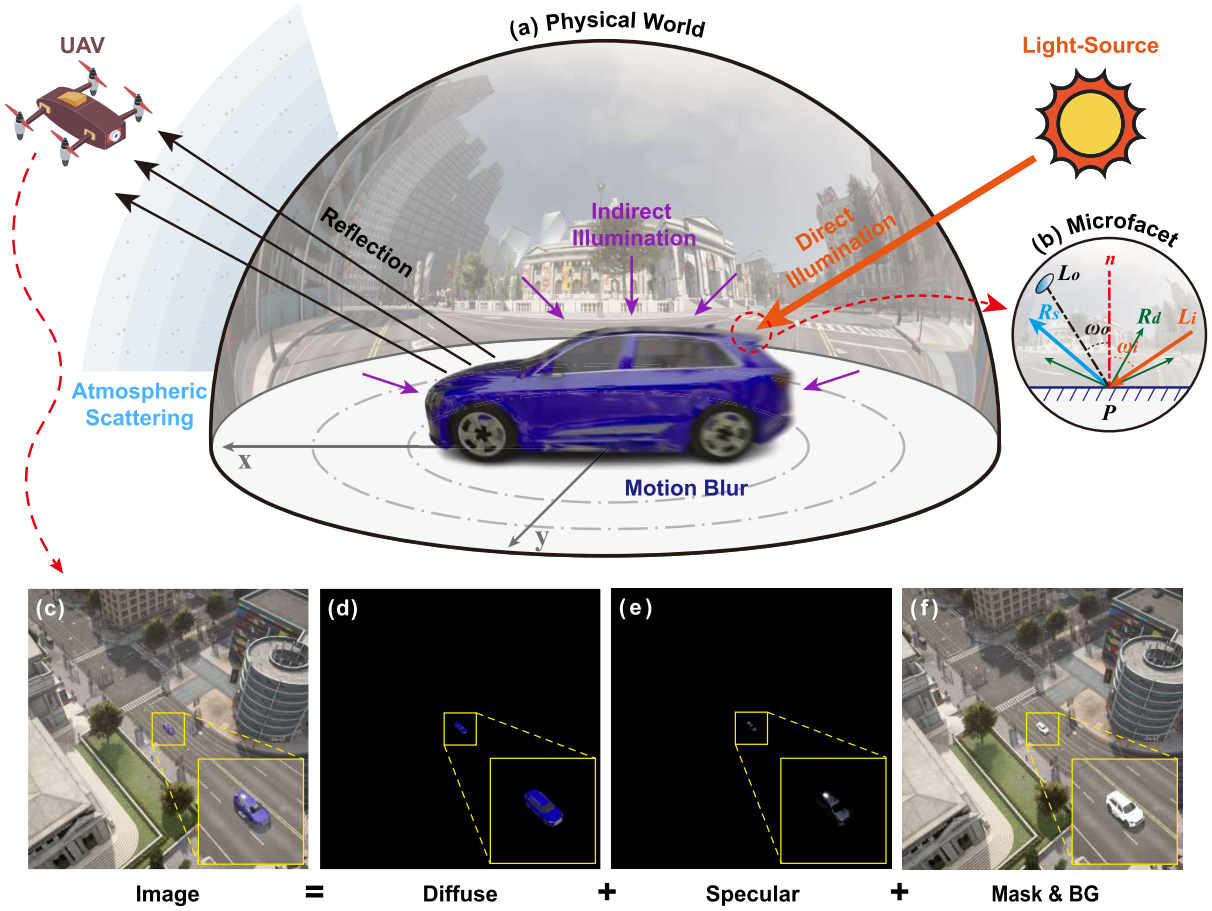


Fig. 3. Physical imaging process in UAV-based detection scenarios, and the rendering and image synthesis mechanism of PBDR. (a) Illumination received by the target in the physical world and its propagation path. (b) Reflection on microsurfaces of the target. (c) Output image rendered and synthesized by PBDR. (d) Diffuse reflection component computed by PBDR. (e) Specular reflection component computed by PBDR. (f) Background and other textureless regions of the target extracted using a mask.

into the texture optimization. In comparison, PBDR precisely computes the irradiance distribution and reflection characteristics of the camouflaged target during training and explicitly incorporates real physical interference factors, such as illumination, motion blur, and atmospheric scattering. This enables the camouflage generation process to jointly consider environmental variations and imaging mechanisms, effectively back-propagating physical perturbations into the texture optimization. As a result, the generated camouflage maintains stable adversarial performance and strong physical robustness under varying viewpoints, lighting conditions, and distances.

To understand the rendering principle of PBDR, we first start from the physical imaging process of the real world [36], [37], [38]. As illustrated in Fig. 3(a), the primary natural light source in UAV surveillance scenarios is typically the sun. The illumination on the target surface consists not only of direct sunlight but also of indirect light scattered from the surrounding environment. After reaching the target, these light rays undergo both specular reflection and diffuse reflection on their surface before being captured by the UAV camera to form an image. Unlike other methods that adopt a simplified illumination model considering only direct lighting, PBDR employs a global illumination model that accounts for both direct and indirect lighting to better reproduce the performance



Fig. 4. Environmental map, irradiance map, and light-source information used in PBDR.

of adversarial camouflage in real-world physical tasks. Moreover, to more faithfully simulate actual UAV-based scenarios, PBDR additionally incorporates atmospheric scattering and motion blur interference during the UAV imaging.

To calculate the reflection phenomenon on the target surface under global illumination, it is necessary to start from the microfacets of the target surface, as illustrated in Fig. 3(b). Here, L_i and L_o denote the incident and outgoing radiance, respectively, ω_i and ω_o represent the incident and outgoing angles, respectively, p is a microfacet on the target surface, n is the facet normal, and R_s and R_d correspond to the specular and diffuse reflection components. The outgoing radiance

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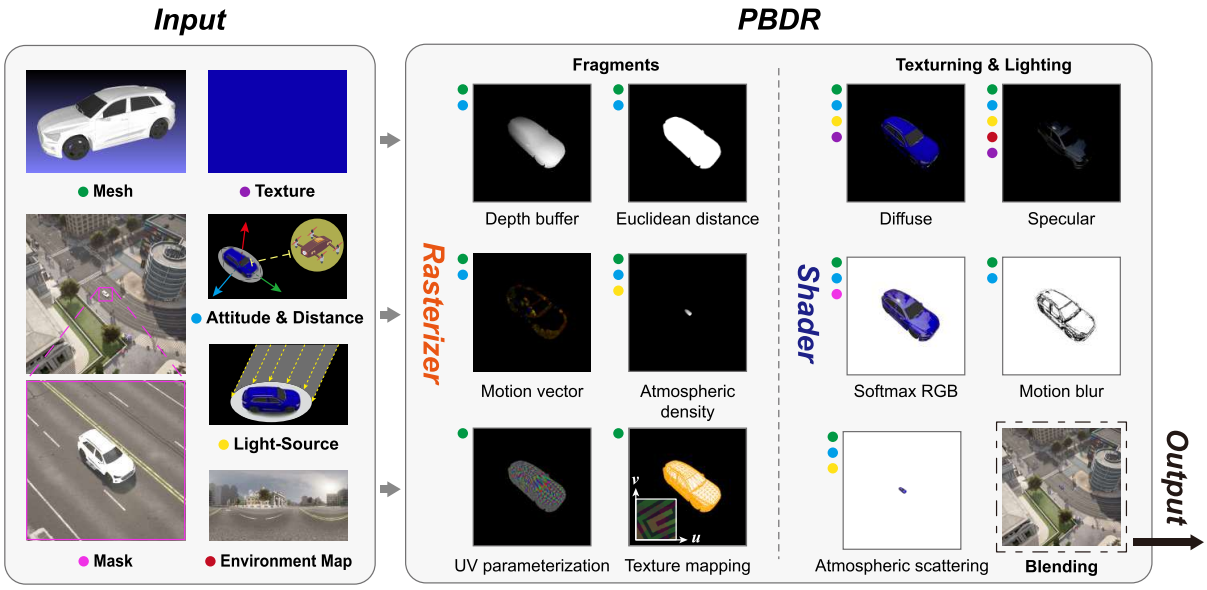


Fig. 5. Rendering pipeline of PBDR, where the differently colored dots adjacent to each step in the rasterizer and shader on the right side correspond to the input data on the left side, indicating which input information is utilized at each step.

contributed by diffuse reflections can be obtained through the integrals in the following equation:

$$L_o^D(p,\omega_o) = k_d \frac{c}{\pi} \int_\Omega L_i^D(p,\omega_i)\, n \cdot \omega_i d\omega_i. \tag{3}$$

Also, the specular reflections can be obtained through the integrals in the following equation:

$$L_o^s(p,\omega_o) = k_s \int_\Omega L_i^s(p,\omega_i)\, \mathrm{d}\omega_i \cdot \int_\Omega f_r(p,\omega_i,\omega_o)\ (n \cdot \omega_i)\, \mathrm{d}\omega_i. \tag{4}$$

Here, k_d and k_s denote the diffuse reflection coefficient and specular reflection coefficient of the target surface material, respectively. In (3), the diffuse reflection of a facet in the outgoing direction ω_o is computed via integration, as shown in Fig. 3(d). In (4), the specular reflection of a facet in the outgoing direction ω_o is also computed via integration, as shown in Fig. 3(e). This integration operation consists of two parts: the first accounts for the specular reflection integral of indirect illumination, and the second accounts for that of direct illumination. For the indirect illumination, the incident light is derived from the environmental map, while for the direct illumination, the incident light is modeled as a directional light source to simulate sunlight, as shown in Fig. 4.

The previous content introduced the physical process modeling of imaging in PBDR. The complete structure and rendering pipeline of PBDR are illustrated in Fig. 5. As shown, for each training iteration, the inputs to PBDR include not only the image but also the mesh model, the attitude and distance of target and camera, the light source position, the environmental map, and the mask. PBDR is composed of two main parts: rasterizer and shader. The rasterizer mainly handles geometric data processing, including vertex transformation, triangle clipping, depth computation, and pixel interpolation, generating geometric information such as the depth buffer, world coordinates, and UV mapping. The shader is responsible for material and lighting calculations, including texture sampling, normal processing, lighting model computation, motion blur, and atmospheric effect simulation. Finally, the shader combines the synthesized color results with the background information extracted from the input image using the mask to produce a new composite image.

For the construction of the global illumination model, PBDR integrates over each surface element of the model along the UAV viewing direction using (3) and (4) to obtain diffuse and specular reflections. Then, in the image color composition stage, multiple candidate color components are weighted and normalized to ensure that the color values remain stable and differentiable during backpropagation.

For the simulation of motion characteristics [39], [40], PBDR first models the rapid variation of the observation viewpoint during target movement, thereby capturing the transient changes in viewing angles caused by the motion. In addition, PBDR generates motion vectors along the theoretical motion direction of the target model (e.g., the vehicle's heading) during the rasterization stage, and computes the motion offset of each pixel in screen space based on the target's rigid-body motion parameters. In the shading stage, the multiframe sampling results are then weighted and blended along the motion trajectories according to the direction and magnitude of the motion vector field, thereby producing a motion blur effect consistent with physical laws.

In UAV imaging, the atmospheric effects that play a significant role include Rayleigh [41] and Mie scattering [42]. However, when the distance between camera and target is only a few hundreds of meters, the color shift caused by Rayleigh scattering is weak and can be neglected. Therefore, in the simulation of atmospheric effects, PBDR mainly considers the contrast reduction effect caused by Mie scattering.

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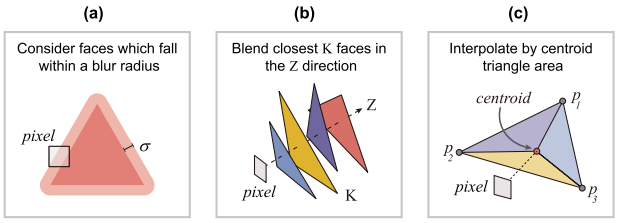


Fig. 6. Strategiesfor dealing with typical discontinuity problems in traditional rendering. Three representative problems along with their corresponding solutions in PBDR. (a) Face boundary. (b) Face occlusion. (c) UV mapping.

Accordingly, in the renderer design, the rasterizer outputs the world-coordinate height value of each pixel as the atmospheric density mask, the shader computes the fog factor based on this height value and depth information using the Mie scattering physical model, as expressed in the following equation:

$$f = 1 - \exp\left(-\beta_m * \rho * d\right). \tag{5}$$

Here, f denotes the Mie scattering fog factor, β_m represents the Mie scattering coefficient, ρ is the atmospheric density, and d is the distance from the observation point to the object. Then, based on the fog factor, the contrast of objects is attenuated to reproduce the visual effects of atmospheric influence in UAV imaging, as shown in the following equation:

$$C_m = C_{\text{object}} * (1 - f) + C_{\text{fog}} * f. \tag{6}$$

Here, C_m denotes the pixel color value after incorporating Mie scattering, C_{obj} represents the object pixel color value obtained from integrating the illumination model, and C_{fog} is the fog color value of Mie scattering (typically grayish white).

In PBDR, when implementing differentiable functionality, a probability-based soft rasterization strategy is employed to address gradient discontinuities caused by face boundaries, face overlaps, and UV mapping discontinuities [43], [44], [45], as shown in Fig. 6. Specifically, first, a differentiable sigmoid function is used to soften geometric boundaries, transforming discrete boundary relations into smooth transitions and ensuring gradient propagation across boundaries. Second, the traditional hard occlusion is replaced with a probabilistic transparency blending approach, where a weighted accumulation method calculates the gradient contributions of overlapping facets, allowing textures in partially occluded regions to receive effective gradient updates. Third, for UV discontinuities, the bilinear interpolation is applied during texture sampling to propagate pixel gradients continuously, preventing abrupt changes in optimization direction at texture seams. These methods ensure stable gradient flow at face boundaries, occluded regions, and texture discontinuities in PBDR, supporting end-to-end differentiable rendering pipeline training.

C. Pyramid Camouflage Training Framework

PCTF is an adversarial texture generation method designed for multiscale detection scenarios. Its core lies in achieving cross-scale consistency optimization of camouflage textures through a distance-layered training mechanism, with the network architecture shown in Fig. 7. Therefore, before feeding data into PCTF for training, the image dataset is first organized into layers according to shooting distance. Each layer corresponds to a specific distance range and contains image samples captured at various angles, lighting conditions, and background environments within that distance.

To establish cross-scale adaptability, training in PCTF begins at long distances. When the target is imaged from a far distance, it occupies fewer pixels, so using low-resolution textures helps the camouflage generation model focus on the contours and low-frequency color features of the texture, avoiding interference from high-frequency noise. After training on the data layer corresponding to the current shooting distance, the resulting texture is upsampled and used as the initial texture for the next, closer distance data layer. This guides the higher resolution texture in the next layer to inherit and refine these texture features. Meanwhile, a multiscale consistency loss is imposed between adjacent layers to preserve the key features that are effective at long distances. This approach balances the richness of details with contour consistency, ultimately producing textures that exhibit scale invariance and strong adversarial capability for multiscale detection tasks on UAVs.

In the layered computation strategy of PCTF, the choice of texture resolution for each layer is crucial. According to the Nyquist sampling theorem [46], to avoid losing details, the texture sampling frequency should be at least twice the highest spatial frequency of the target in the image. If the texture resolution is too low, the available pixel area for applying perturbations becomes insufficient, resulting in adversarial texture patterns that are overly simple and coarse, lacking sufficient detailed features to effectively deceive the detector's complex feature extraction networks, thereby significantly weakening the camouflage's attack effectiveness. Conversely, if the texture resolution is too high, unnecessary oversampling occurs, which not only greatly increases computational overhead but also risks erroneously smoothing out the high-frequency details added by the camouflage generation network during rendering. The design formula for the texture resolution of each layer in PCTF is given in the following equation:

$$R_{\text{tex}} = 2 \cdot \alpha \cdot \sqrt{A}. \tag{7}$$

Here, R_{tex} denotes the texture resolution of the current layer, and α is the supersampling factor used to compensate for area differences caused by partial visibility, surface deformation, or nonfrontal views. In our work, $\alpha = 2$, A represents the number of pixels occupied by the target in the image at the current shooting distance. Typically, for UAV-captured images with a resolution of 3840×2160, when the shooting distance is 50 m, the target occupies approximately 4000 pixels in the image, resulting in $R_{\text{tex}} \approx 256$.

To establish multiangle adaptability, PCTF introduces a multiangle optimization strategy within each layer of training. A random sampling mechanism is employed to select input images from different angles, ensuring coverage across azimuth angles from 0° to 360° and elevation angles from 50° to 90°.

In addition, during the multiangle data training within each layer, the most critical issue is the self-occlusion of the target.

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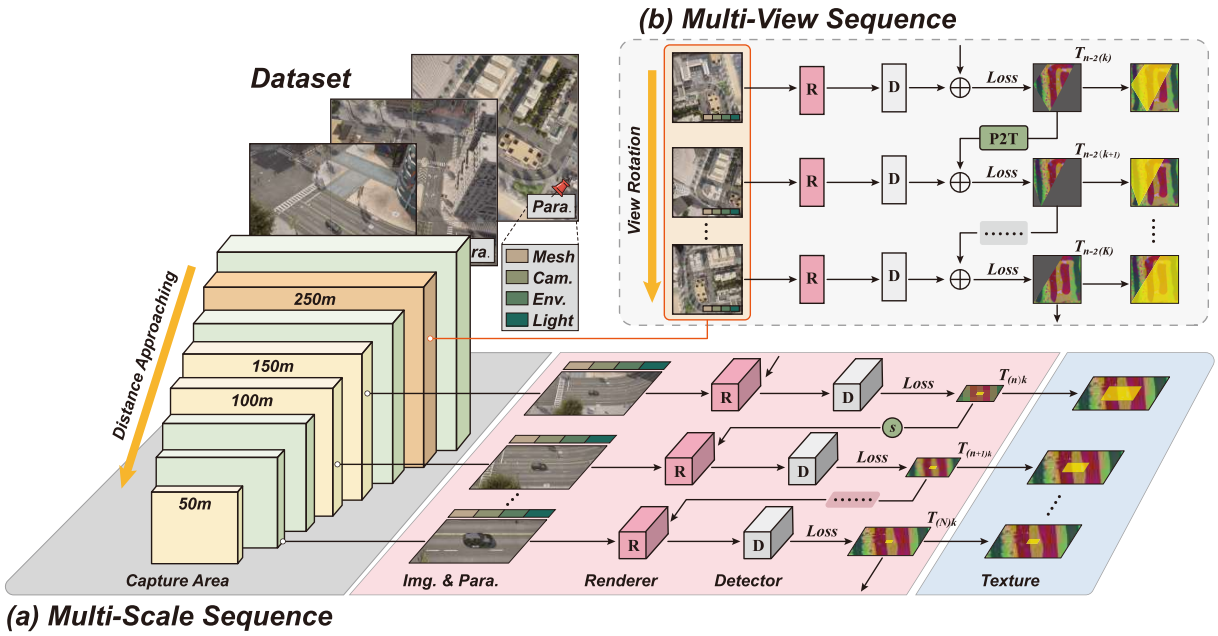


Fig. 7. Structure of PCTF. (a) Interlayer training procedure based on multiscale data. (b) Intralayer training procedure based on multiangle data.

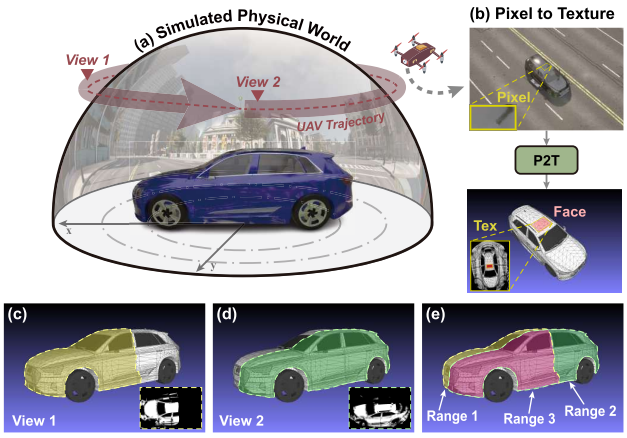


Fig. 8. Schematic of the P2T module. (a) Different shooting positions during UAV cruising, where view1 is the preceding frame of view2 in the training process. (b) Backwards mapping relationship in the P2T module, from image pixels to model faces and then to texture maps. (c) Visible parts of the target under view1. (d) Visible parts of the target under view2. (e) Differences in visible parts between the two frames, where range1 denotes parts visible only in view1, range2 denotes parts visible only in view2, and range3 denotes the parts visible in both frames.

From any viewing angle, a UAV cannot capture all surfaces of the target in a single image; some parts are always occluded and do not appear in the captured image, as shown in Fig. 8(a). In our loss design, the adversarial loss $\mathcal{L}_{\text{ad}}$ is computed based on images synthesized via PBDR rendering, while the color loss $\mathcal{L}_{\text{nps}}$ and distance loss $\mathcal{L}_{\text{dis}}$ are computed based on the target model's texture map. Consequently, in each iteration, the visible parts of the target in the image are constrained by all three losses, whereas the occluded parts are only constrained by $\mathcal{L}_{\text{nps}}$ and $\mathcal{L}_{\text{dis}}$. This self-occlusion phenomenon causes occluded regions to gradually degrade into uniform-colored areas due to the lack of adversarial supervision. This effect is particularly pronounced in long-distance layers, where the sides of targets (e.g., car doors) are often only partially visible; after processing in such layers, the side textures quickly degenerate into uniform colors, preventing these regions from being properly optimized in subsequent layers.

To address this issue, we design a pixel-to-texture (P2T) module in PCTF, as illustrated in Fig. 8(b). This module uses PBDR to back-project the image regions where the target is visible onto the corresponding texture areas. For both the current view [see Fig. 8(c)] and the previous view [see Fig. 8(d)], the texture regions corresponding to the visible parts of the target, denoted by range1 and range2, are computed. A visibility-based activation matrix is then established to activate textures visible in the current view and suppress those that are not. Only the texels that are activated are updated in the current iteration, while suppressed texels retain their state from the previous iteration, as shown in Fig. 8(e). Notably, during the output stage of this iteration, texture regions that are activated in both the current and previous iterations (range3) are combined through summation and averaging rather than being directly updated. This further ensures interframe consistency across multiple viewing angles in the camouflage generation.

The above describes the framework design of PCTF. Equally important is the loss function configuration during the PCTF training process. In our work, three loss functions are used, as shown in the following equation:

$$\mathcal{L} = \mathcal{L}_{\text{ad}} + \mathcal{L}_{\text{dis}} + \mathcal{L}_{\text{nps}}. \tag{8}$$

1) The design of the adversarial loss $\mathcal{L}_{\text{ad}}$ is shown in the following equation:

$$\mathcal{L}_{\text{ad}} = \text{iou} + \text{cls} + \text{obj}. \tag{9}$$

Here, iou represents the intersection-over-union between the model's predicted region on the rendered image

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and the corresponding ground-truth region, cls denotes the classification score for predicting the target in the rendered image as the correct class, and obj is the confidence score indicating whether the model believes a target exists in the rotated bounding box. During training, the constraints imposed by $\mathcal{L}_{\text{ad}}$ gradually reduce the values of these three components. A decrease in iou reduces the overlap between the predicted and ground-truth bounding boxes, making it harder to accurately localize the camouflaged target. A decrease in cls lowers the model's confidence in the correct class, enhancing the camouflage's ability to confuse classification. A decrease in obj directly weakens the detector's confidence in the target's existence, increasing the likelihood of inducing missed detections.

2) The design of the scale loss $\mathcal{L}_{\text{dis}}$ is shown in the following equation:

$$\mathcal{L}_{\text{dis}} = F\left(S_d\left(T_n\right)\right) - F\left(T_{n-1}\right). \tag{10}$$

Here, F is a pretrained VGG network used to extract texture features, S_d denotes the downsampling operation applied to the texture and T_n and T_{n-1} represent the textures of the current and previous layers, respectively. In the layered training of PCTF, the constraints imposed by $\mathcal{L}_{\text{dis}}$ preserve the key features in the textures at longer distances and refine the texture details through progressively closer distance image training, achieving a balance between rich detail and contour consistency.

3) The design of the printable loss $\mathcal{L}_{\text{nps}}$ [47] is shown in the following equation:

$$\mathcal{L}_{\text{nps}} = \sum_{ij} \min_{c_{\text{print}} \in C} \left| p_{ij} - c_{\text{print}} \right|. \tag{11}$$

This loss controls the color of each texel p_{ij} in the adversarial texture by measuring its deviation from a predefined color set c_{print}, encouraging the texels to match the colors in the set closely. In our work, the eight colors in c_{print} are determined by performing k-means clustering on the output textures from multiple experiments without color constraints. It is also worth noting that when the textures are printed into the physical world, the printer uses the CMYK color space, whereas training is typically conducted in the RGB color space. Therefore, the selected colors in c_{print} are converted to CMYK to reduce the digital-to-physical discrepancy caused by the printing process.

Then, by taking the derivative of the total loss function $\mathcal{L}$ with respect to the texture T, the gradient signal g_k is obtained and propagated through the inverse rendering process to the surface texture of the target model, thereby enabling iterative updates of the camouflage texture under multiple constraints. To improve the training stability and convergence speed, this work adopts the adaptive moment estimation optimizer, which dynamically adjusts the learning rate of each parameter based on the first and second moments of the loss function gradient.

First, the first-order and second-order moment estimates of the gradient are computed as follows:

$$\begin{cases} m_k = \beta_1 m_{k-1} + (1-\beta_1)\, g_k \\ v_k = \beta_2 v_{k-1} + (1-\beta_2)\, g_k^2 \end{cases} \tag{12}$$

where β_1 and β_2 denote the exponential decay rates for the first and second moment estimates, respectively. To eliminate the initialization bias, the bias correction is applied to m_k and v_k

$$\begin{cases} \hat{m}_k = \dfrac{m_k}{1-\beta_1^k} \\ \hat{v}_t = \dfrac{v_k}{1-\beta_2^k}. \end{cases} \tag{13}$$

Finally, the model parameters are updated based on the bias-corrected gradient

$$T_{k+1} = T_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon} \tag{14}$$

where α is the learning rate and ϵ is a small constant to prevent division by 0.

The overall method is shown in Algorithm 1.

Algorithm 1 Adversarial Camouflage Generation

Input: mesh $\mathbf{M}$, training set ($\mathbf{I}$, $\mathbf{\Phi_v}$, $\mathbf{\Phi_d}$, $\mathbf{\Phi_l}$, $\mathbf{m}$), PBDR model $\mathbf{R}$, detector $\mathbf{D}$
Output: Adversarial texture T

```
Initialize T with random noise;
for the number of epochs do
    Sample minibatch
      I ∈ I, θ_v ∈ Φ_v, θ_d ∈ Φ_d, θ_l ∈ Φ_l, m ∈ m;
    for the number of epochs do
        I_adv ← m · R(M, T, θ_v, θ_d, θ_l) + (1 − m) · I;
        O ← D(I_adv);
        Calculate loss L by (8);
        Update T for minimizing loss L via back
          propagation and F2T model;
        Upsample T to the resolution of the next layer;
    end
end
return T;
```

IV. Experimental Results

In this section, we conduct the comprehensive and extensive experiments to demonstrate the effectiveness of our proposed method. First, we present the relevant experimental settings (see Section IV-A) and evaluation indicators (see Section IV-B). Next, we perform comprehensive experiments in both the simulated (see Section IV-C) and the real physical world (see Section IV-D), providing a detailed explanation and analysis of the specific experiments and results. Finally, we discuss the role of each loss function and functional module in our method through ablation studies (see Section IV-E).

A. Experimental Settings

1) Dataset: We adopt the CARLA simulator to generate the dataset to better compare with previous works and simulate

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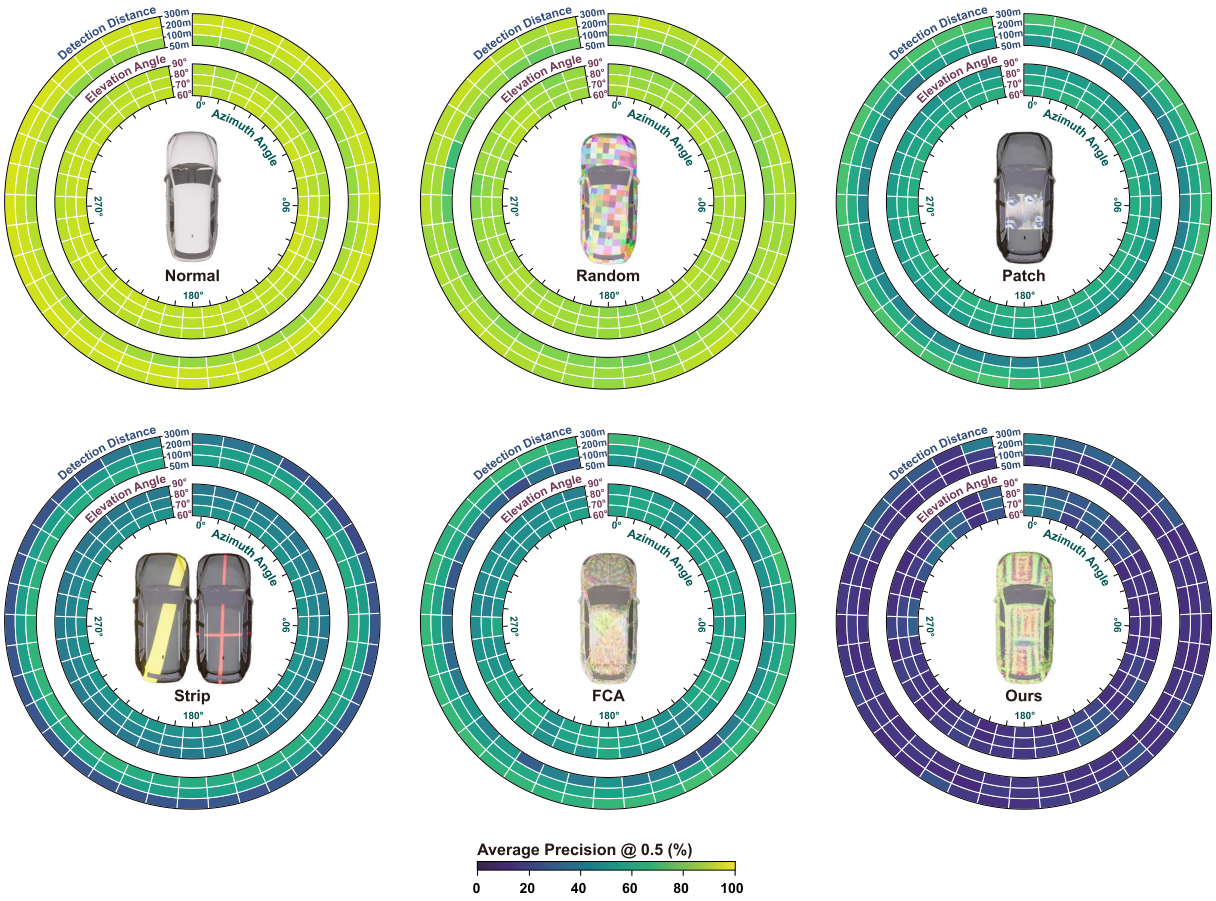


Fig. 9. Results of experiments in the simulated physical world, where six sets of circular diagrams are presented, the top view at the center of each circle indicates the orientation of the model. In the outer ring, the segments from outside to inside represent distances ranging from 300 to 50 m, while in the inner ring, they represent camera elevation angles from 90° to 60°. Each cell within the ring corresponds to a camera azimuth angle, and the values shown are the AP under that condition. Brighter colors indicate higher AP values, which means weaker camouflage performance under that condition, while darker colors correspond to lower AP values, indicating stronger camouflage performance. For the strip method, we count the test results of 1- and 2-Strip camouflage.

real UAV imaging. The 40 different locations are selected across multiple maps in CARLA, with 30 locations used for training adversarial textures and ten locations reserved for testing adversarial camouflage performance. For each location, four lighting conditions are considered: overcast at noon, overcast at dawn or dusk, sunny at noon, and sunny at dawn or dusk. The shooting distances are set to [50, 75, 100, 150, 200, 250, 300] m. The camera elevation angles are set to [50°, 60°, 70°, 80°, 90°] (the detection accuracy of UAV-based detection models for targets within the 0°–50° range is relatively low), and the camera azimuth angles are set to 0°–360°. The image resolution is 3840×2160.

2) Detectors: To generate adversarial camouflage, we first train the camouflage using a white-box model and then evaluate its performance on white-box and black-box models. To ensure fair comparison with prior works, we adopt detector choices consistent with the baseline methods as much as possible. Specifically, in this study, YOLOv5OBB [48] is used as the white-box detector, while Oriented R-CNN [49], S2Anet [50], RetinaNet OBB [51], Faster R-CNN OBB [52], and RoI Transformer [53] are selected as black-box detectors.

3) Baseline Methods: To comprehensively compare the characteristics of different types of camouflage and fully demonstrate the superiority of our proposed method, we select one representative approach for each camouflage type: patch-based, stripe-based, and fully covered camouflage. In addition, we include two extra baselines: a target without camouflage and a target with randomly generated noise textures.

4) Implementation Details: The maximum epoch is set to 10, and the Adam optimizer is used with a learning rate of 0.01. The resolutions for the hierarchical textures are chosen as [32×32, 52×52, 64×64, 90×90, 128×128, 180×180, 256×256].

B. Evaluation Indicators

In physical adversarial attack methods, the indicators commonly used to evaluate the adversarial effectiveness of camouflage are ASR and AP. AP generally refers to the degree to which adversarial camouflage reduces detection accuracy. In our work, we use mAP@0.5, i.e., the average precision (AP) when the IoU between the detected bounding box and the ground truth is 0.5. The lower the AP, the stronger the adversarial effectiveness. ASR represents the probability of successfully evading model detection, calculated as the ratio of successful attacks to the total number of attack attempts. In these experiments, an attack is considered successful only if the detector's confidence score falls below a threshold of 0.5.

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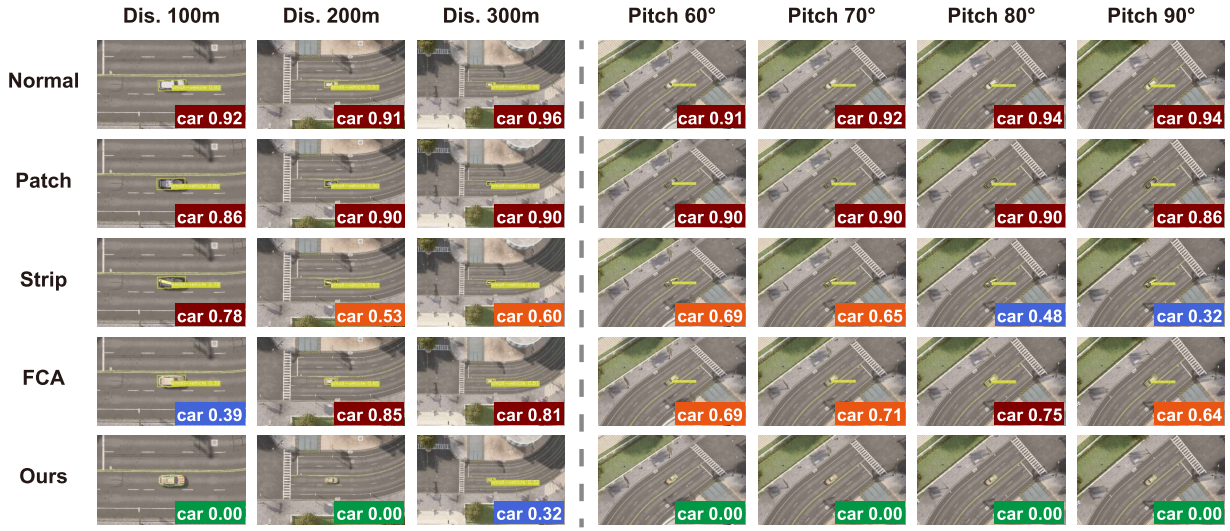


Fig. 10. Typical samples of experimental results for camouflages generated by different methods under various conditions. (Left) Results under different distance conditions. (Right) Results under different camera elevation angle conditions. The labels in the lower right corner indicate adversarial performance levels: red represents almost no adversarial effectiveness (75% < conf ≤ 100%), orange represents weak adversarial effectiveness (50% < conf ≤ 75%), blue represents strong an adversarial effectiveness (0% < conf ≤ 50%), and green represents the most substantial adversarial effectiveness (conf = 0%).

From the perspective of their definitions, both AP and ASR can only measure the ability of camouflage to mislead detection models in static scenarios, but cannot evaluate its sustained misleading capability in dynamic scenarios. Meanwhile, from the perspective of real task requirements, a single correct prediction across multiple consecutive frames is generally considered a successful detection. However, single-frame indicators such as AP or ASR, which are based on averaging, overlook the temporal continuity of object detection and further widen the gap between laboratory results and real-world deployment of physical adversarial attacks. Therefore, we propose temporal ASR (TASR), which quantifies the sustained adversarial capability of camouflage by calculating the proportion of consecutive frames in which the detection model is continuously misled. The definition of TASR is given in the following equation:

$$\mathrm{TASR}_t = \frac{\sum F\,(k > t)}{N}. \tag{15}$$

Here, N denotes the total number of test frames, t is the threshold for consecutive misleading frames (e.g., $t = 10$ means that at least ten consecutive frames must be misled), and $F(k > t)$ represents the total number of frames in the test results that are part of sequences where the consecutive misleading length exceeds t.

C. Comparative Experiments in Simulated Physical World

To validate the proposed method, we conduct comparative experiments in a simulated physical environment. In the experiments, ten different positions were selected, while other conditions, such as viewing angles and illumination, were kept consistent with the training setup. Moreover, we employed YOLOv5OBB in this experiment to evaluate the performance of adversarial camouflages generated by various methods. Fig. 9 illustrates the AP values of the attacked detection model under different viewing angles and distances. Figs. 10 and 11 present representative samples of the experimental results under various angles, distances, and illumination conditions, while Fig. 12 shows the variation trends of the experimental results under different conditions.

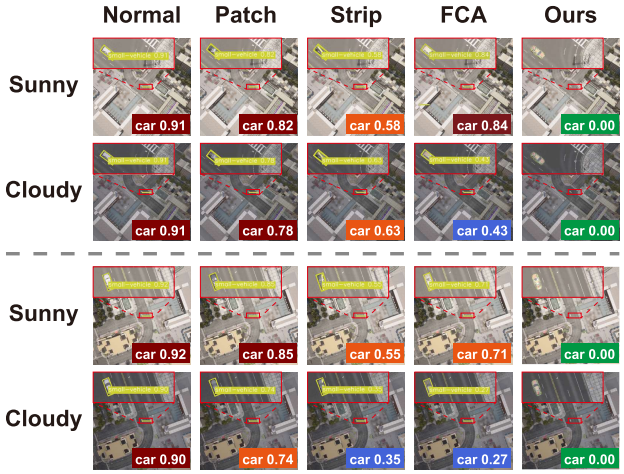


Fig. 11. Typical samples of experimental results for camouflages generated by different methods under various illumination conditions.

1) Different Viewing Angles: First, to better compare the performance differences of various methods under different viewing angles, we categorize the experimental data according to elevation and azimuth angles, as shown in the inner ring of Fig. 9. In addition, the performance trends of different methods across viewing angles are illustrated in Fig. 12(b). The experimental results show that the patch and strip methods perform well when the UAV's elevation angle is 90° (directly overhead). However, as the UAV's elevation angle gradually decreases, the performance of patch and strip declines due to texture distortion caused by the changing viewpoint. Regarding trend analysis, at elevation angles of 90° and 50°, AP of patch increases by 15% and that of strip increases by 16%, while FCA method [5] (4%) and our method (2%) show almost no variation.

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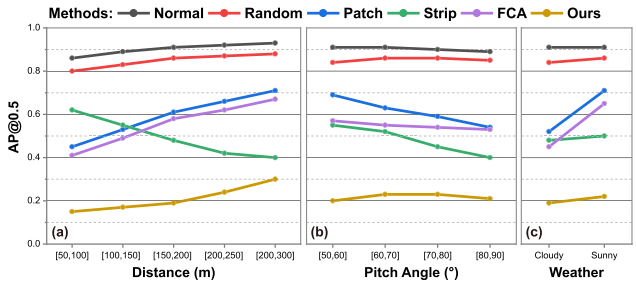


Fig. 12. Line charts of the results from the adversarial attack experiment in the simulated physical world. (a) Results under different distances. (b) Results under different elevation angles. (c) Results under different lighting conditions.

Furthermore, on the right side of Fig. 10, we present representative examples under different camera elevation angles. Due to the inherent adversarial camouflage generation mechanism of their methods, the patch and strip approaches can only be applied to flat regions of the target surface (mainly the roof and the hood). As the UAV's elevation angle decreases, the proportion of the roof and hood in the target region of the image becomes smaller, and more uncovered areas (such as the car doors) are captured by the UAV, which inevitably leads to a decline in camouflage performance. These results, on the one hand, demonstrate that fully covered camouflages are more advantageous than patch- and strip-based methods when confronting multiangle detection from UAVs. On the other hand, they also indicate that incorporating viewpoint variations into the camouflage training process through inverse rendering can significantly enhance the multiangle robustness of camouflage.

2) Different Detection Distances: In adversarial attack studies targeting UAV-based detection tasks, the detection distance is an important factor that most existing methods have failed to consider. Therefore, to compare the performance differences of various methods at different distances, we also recorded experimental results at different distances. The results are shown in the outer ring of Fig. 9, while the performance trend variations across different distances are illustrated in Fig. 12(a).

When the UAV captures images at a closer distance, the performance of strip degrades (AP increases by 22%). This is because the method only considers texture preservation in long-distance scenarios, without accounting for the fact that, as the UAV approaches, it gradually captures finer details of the target. As a result, the method performs worse at near distances than at long distances.

Moreover, as the UAV moves farther from the target, the other two methods show a certain degree of performance degradation. Specifically, AP of FCA and patch increases by 26% and 27%, respectively, while our method only increases by 9%. This is because FCA and patch methods do not incorporate a camouflage optimization mechanism for preserving texture features under long-distance scenarios. In contrast, our method effectively suppresses the degradation of camouflage textures with increasing distance through PCTF.

Similarly, on the left side of Fig. 10, we present typical examples at different distances. It can be observed that at near distances, target details are more clearly visible, and simple uniform colors or vertical stripes in the strip method are insufficient to hide these details from the detector, resulting in degraded camouflage performance. At longer distances, the textures of FCA and strip degrade significantly, appearing almost uniform at 300 m, which causes these two methods to lose nearly all adversarial effectiveness under long-distance observation. In contrast, our method, due to the reproduction of physical conditions at various shooting distances in PBDR and the multiscale training with feature preservation mechanisms in PCTF, demonstrates superior performance at both short and long distances compared with existing methods.

3) Different Illumination Conditions: Apart from viewing angle and distance, the major difference between the digital space and the physical world lies in the illumination model. Strip and patch do not build the illumination model, while the lighting model used in FCA is overly simplistic, considering only direct illumination and diffuse reflection. To compare the performance differences of various methods under different illumination conditions and to analyze the influence of illumination models on adversarial camouflage, we also evaluate the performance trends of each method under varying illumination. The statistical results are presented in Fig. 12(c), with two representative examples in Fig. 11.

It can be observed that patch and FCA are highly sensitive to illumination conditions. When the illumination is weak (e.g., on cloudy days), their performance is closer to that observed in controlled laboratory settings. However, under strong lighting, the texture color values of the camouflage are subject to larger shifts, and specular highlights often appear that directly occlude the textures. These factors collectively result in poor robustness of patch and FCA under complex natural illumination. In contrast, our method incorporates accurate physical interference factors into the adversarial camouflage training process through global illumination modeling in PBDR, thus achieving stronger robustness against complex real-world lighting conditions and specular reflection occlusions.

It is also worth noting that strip is relatively less affected by illumination, mainly due to its design with limited color variation and directional consistency. This avoids the color temperature sensitivity of high-frequency texture details and allows it to maintain relatively stable contrast in environments with strong illumination and color bias. However, these same design choices also make strip camouflage relatively simple, leading to performance degradation under multiangle and near-distance conditions.

4) Dynamic Scenarios: As mentioned earlier, UAV-based detection is predominantly conducted in video, where conventional video typically operates at 20~30 frames/s, while in tasks requiring higher accuracy, the frame rate may be reduced to 5–10 frames/s. In such tasks, the detection is considered successful if at least one frame within a second successfully identifies the target.

To compare the performance differences of various methods in dynamic scenarios, we conducted two sets of experiments in the simulator. In the first set, each camouflaged target was assigned the same path and tested individually. In the second set, all camouflaged targets were tested simultaneously in

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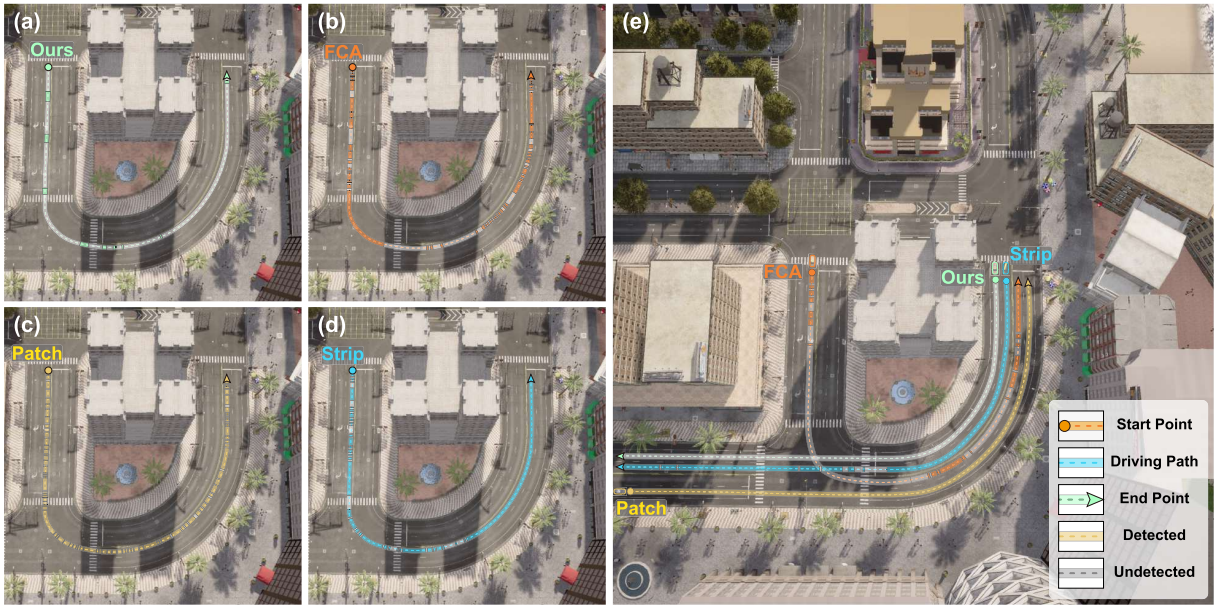


Fig. 13. Experimental results in dynamic scenarios. (a)–(d) Detection results of camouflaged targets generated by different methods along the same trajectory. (e) Detection results of all camouflaged targets in the same scene.

the same scene. The experimental conditions were set with a shooting distance of 200 m, a target speed of 60 km/h, and a confidence threshold of 50% for the detection model. The results are shown in Fig. 13, where circles indicate path starting points, arrows indicate endpoints, and different colors correspond to the trajectories of targets camouflaged by different methods. In addition, bright segments represent detected positions, while faint segments represent undetected positions.

Fig. 13(a)–(d) corresponds to the same-path experiments. It can be observed that our method achieves stronger sustained adversarial capability under dynamic conditions, benefiting from the motion characteristics simulation in PBDR and the interframe feature preservation mechanism in PCTF. The next strongest performance is observed with FCA, which corroborates the earlier analysis: although none of the methods explicitly consider dynamic scenarios, the advantage of full-covered camouflage under multiangle detection grants FCA greater robustness in dynamic settings compared with the other approaches. Strip ranks third in terms of sustainability, although it performs better than FCA in some static viewpoints, its inherent design limitations lead to weaker performance in dynamic conditions compared with FCA. Patch ranks the lowest, partly because its baseline capability is inherently weaker than both FCA and strip.

Fig. 13(e) corresponds to the same-scene experiment. In this case, our method achieves complete evasion of detection throughout the entire experimental duration, effectively realizing the visual invisibility in machine vision. The stability of other methods remains consistent with their performance in the same-path experiment, further validating our analysis. It is worth emphasizing that we deliberately adjusted the lighting to create distinct shadowed and illuminated regions in the scene. As observed, strip is largely insensitive to lighting, indicating that its limited hue variation and directional consistency help mitigate the color temperature sensitivity of high-frequency texture details. In contrast, FCA performs significantly better in shadowed areas than in illuminated areas, showing strong sensitivity. Moreover, the texture shifts of FCA under strong lighting could be directly observed in the video. These results once again corroborate our earlier analysis.

TABLE I
STATISTICS OF EXPERIMENTAL RESULTS IN DYNAMIC SCENARIOS (%)

Method	ASR	$TASR_5$	$TASR_{10}$	$TASR_{20}$	$TASR_{30}$
Patch	34.91	19.17	17.47	11.18	7.35
Strip	44.69	24.20	21.42	12.92	8.99
FCA	53.30	48.21	44.66	40.62	38.89
Ours	**88.58**	**87.19**	**86.11**	**84.26**	**80.40**

In addition, to quantify the sustained adversarial capability of each method, we calculated the ASR and TASR of the camouflaged targets in the dynamic experiments, as shown in Table I. The results indicate that when considering TASR, even within just 5 frames ($TASR_5$), the score of patch and strip drops significantly by 45.1% and 31.0%, respectively, while FCA only decreases by 9.6%, showing a smaller decline than the other two. In contrast, our method experiences merely a 1.6% reduction. Furthermore, when the threshold of TASR exceeds 20 frames, the performance of patch and strip sharply decreases to below 10%, implying that these methods are essentially ineffective in UAV-based dynamic detection tasks (typically set at 30 frames/s). Although FCA maintains some effectiveness at 30 frames, its performance decreases by 27.0%. In comparison, our method achieves an average of 84.5% from $TASR_5$ to $TASR_{30}$, with a maximum drop of only 9.2% at 30 frames. Even at 60 frames, $TASR_{60}$ of our method remains as high as 57.87%. These results demonstrate

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TABLE II

ADVERSARIAL ATTACK PERFORMANCE COMPARISON BASED ON WHITE-BOX AND BLACK-BOX MODEL TESTS, VALUES IN THE TABLE ARE AP (%) AND ASR (%), RESPECTIVELY

	YOLOv5-OBB		Oriented R-CNN		S2ANet		RetinaNet OBB		Faster RCNN OBB		Rol Transformer	
	AP	ASR	AP	ASR	AP	ASR	AP	ASR	AP	ASR	AP	ASR
Normal	90.16	-	85.56	-	83.52	-	92.04	-	85.00	-	94.44	-
Patch	59.43	44.70	72.55	27.04	68.42	25.19	70.62	32.04	72.58	22.41	63.70	39.07
Strip	48.01	53.70	69.28	38.70	60.30	32.41	59.98	31.48	79.34	26.85	58.86	44.44
FCA	52.09	46.07	64.39	39.67	62.74	32.41	65.05	33.89	68.63	38.52	67.29	37.78
Ours	**21.85**	**89.93**	**40.76**	**68.70**	**48.06**	**59.26**	**49.18**	**56.30**	**56.42**	**46.68**	**48.23**	**54.07**

that our approach is better-suited for UAV dynamic detection tasks, whether in standard recognition scenarios (30 frames/s) or highly dynamic scenarios (60 frames/s).

5) Different Detection Models: To more comprehensively evaluate our method, in addition to the white-box attack experiments based on YOLOv5OBB described above, we also conducted black-box attack experiments, with the results shown in Table II. Compared with the white-box results, the performance of camouflage generated by all methods declines in black-box experiments.

However, our method still maintains more than half of the samples, achieving successful attacks. Furthermore, in terms of the average performance across methods, our approach achieves an average ASR of 57% against black-box models, and AP of the camouflaged target drops by an average of 40% compared with the normal target during the experiments. In contrast, other methods yield an average ASR of only 29%~37%, with the corresponding AP drop averaging 18%~23%. Compared with these baselines, the AP score of our method decreases by 18.4% on average, while ASR increases by 70.4%.

These results demonstrate that, although we did not explicitly incorporate considerations for multimodel attacks, merely enhancing the robustness of adversarial camouflage under multiangle and multiscale detection, complex illumination conditions, and dynamic scenarios already enables the camouflage generated by our method to surpass the baseline attack performance of other approaches.

6) Comprehensive Comparison: After conducting the preceding experiments, besides comprehensive adversarial performance (CAP), we also performed statistics and analysis on multiview robustness (MVR), multiscale robustness (MSR), complex illumination robustness (CIR), and dynamic scene robustness (DSR) of these methods. The average ASR of camouflaged targets across static and dynamic experiments measures the CAP. The variance of ASR quantifies MVR, MSR, and CIR under different viewing angles, detection distances, and illumination conditions. DSR is represented by $TASR_{5:30}$ (i.e., the average score from $TASR_5$ to $TASR_{30}$). The robustness indicators derived from variance were further normalized to make the comparison more straightforward. The final comparison results are shown in Fig. 14.

It is evident that, first, our method outperforms existing approaches across all aspects, especially in terms of MSR and DSR; this performance is largely attributed to PBDR and PCTF. Second, these results further validate our earlier analysis: full-covered camouflage achieves better MVR than patch-based or stripe-based methods, and high-frequency texture details are highly sensitive to illumination.

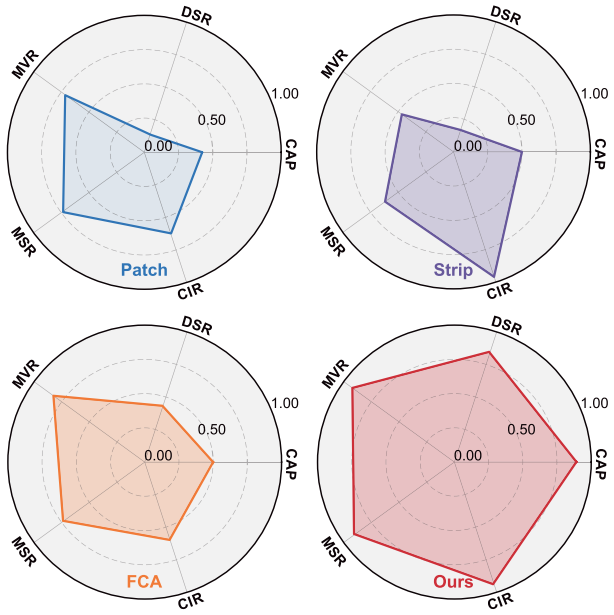


Fig. 14. Comprehensive comparison between methods in adversarial performance.

TABLE III

COMPUTATIONAL EFFICIENCY COMPARISON BETWEEN PBDR AND OTHER TYPICAL DIFFERENTIABLE RENDERERS

Renderer	Method	Time Per Frame
NMR	FCA, DAS	1104ms
PyTorch 3D	ASA	563ms
PBDR	Ours	1441ms

In terms of computational efficiency, compared with conventional training strategies, the additional computational cost introduced by PCTF is not significant. Although texture resolution conversion is required when computing the loss function, the use of low-resolution textures at longer distances effectively reduces the overall computational burden. Moreover, the pixel–face–texture mapping relationships used in the F2T module are inherently recorded in the fragments during the forward rendering process of PBDR, making the additional computational overhead negligible.

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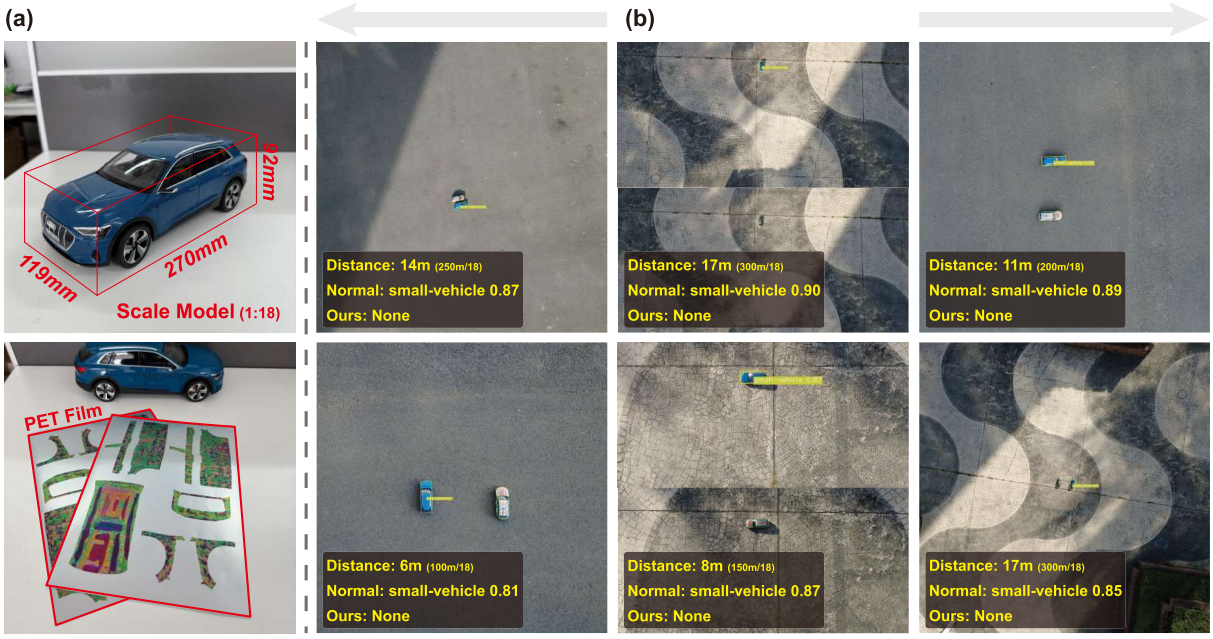


Fig. 15. Physical experiment setup and typical samples of experimental results. (a) Experimental setup. (b) Experimental results.

However, compared with other renderers, PBDR incorporates a more comprehensive set of physical factors, which inevitably leads to higher computational costs. The computational efficiency of PBDR is compared with that of the differentiable renderers used in other methods. In the experiment, the input model contains 23 145 faces, and the rendering is performed on an NVIDIA A100 GPU. The average time consumption per frame during the iterative process is shown in Table III.

D. Physical Adversarial Attack

In order to evaluate the real-world performance of the proposed method and support the previous analysis, we printed the camouflage on PET film and attached it to a 1:18 scale car model for physical-world experiments. The experimental setup is shown in Fig. 15(a). The body of the car model is made of the same alloy material as the real vehicle and manufactured using the original Audi production process, accurately reproducing the target's surface structural features. The PET film, serving as the camouflage carrier, has strong extensibility to conform to the model's surface contours, and its aluminized coating ensures reflective properties similar to those of a real car cover. In addition, a lightweight UAV was used to capture images during the experiments. The UAV is equipped with a 12-Mpixel CMOS sensor and a wide-angle lens (maximum field of view 83°, equivalent focal length 24 mm, aperture $f/2.8$, 1.5-m pixel size, and 1/2.3″ sensor size).

During image acquisition, in order to comprehensively evaluate different physical-world detection angles, the elevation angles were sampled at intervals of approximately 20°, the azimuth angles at intervals of about 45°, and the detection distances were scaled from 50~300 down to 3~17 m according to the 1:18 model scale. For the shooting environment, experiments were conducted at multiple locations with different backgrounds (such as driveway asphalt pavement, or off-highway urban pavement) and under both sunny and cloudy weather conditions, resulting in more than 300 valid images. A selection of results under various elevation angles, azimuth angles, locations, distances, weather, and lighting conditions is shown in Fig. 15(b). It can be clearly observed that the proposed method adapts well to complex disturbances in the physical world while maintaining strong adversarial performance. Moreover, a dynamic experiments were also conducted in the physical world, as shown in Fig. 16, demonstrating that our method is able to retain high adversarial effectiveness even

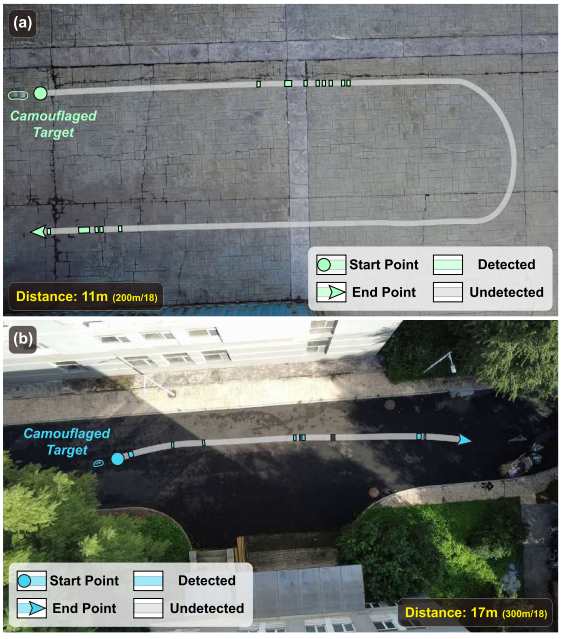


Fig. 16. Experimental results of physical adversarial attacks in dynamic scenarios. (a) and (b) Two sets of experimental results at different distances.

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TABLE IV
STATISTICS OF THE PHYSICAL EXPERIMENTAL RESULTS (%)

	AP	ASR	$TASR_{5:30}$
Normal	87.61	-	-
Ours	24.43	85.96	75.27

in scenarios where relative motion exists between the UAV and the camouflaged target.

Finally, as shown in Table IV, the results of the physical adversarial attack experiments are presented. Under the same viewing angle and scaled distance settings, the AP and ASR scores are largely consistent with those obtained in the simulated physical environment, with differences of approximately 12.3% in AP and 5.1% in ASR. Under the same illumination conditions, the differences in shaded regions are 11.7% for AP and 4.9% for ASR, while in strongly illuminated regions, the differences are 12.9% for AP and 5.3% for ASR. In contrast, other studies reported a digital-to-physical gap of 103.5% in AP for FCA, 50% in ASR for patch, and about 13.8% in ASR for strip. These results demonstrate that, with sufficient consideration of physical-world interference factors, our method achieves better physical robustness compared to other approaches.



Fig. 17. Typical experimental results of adversarial performance testing after transferring our camouflage to different types of vehicles. Zoom for detail.

E. Ablation Studies

In this subsection, we investigate the impact of different loss functions and functional modules on the performance of adversarial camouflage. The results of the ablation study are presented in Table V.

Through the ablation study, we find that both $\mathcal{L}_{\text{dis}}$ and $\mathcal{L}_{\text{nps}}$ contribute to improving the adversarial performance of camouflage. Specifically, $\mathcal{L}_{\text{dis}}$ enhances MSR by preserving texture features across multiscale detections, thereby improving the adversarial effectiveness. The role of $\mathcal{L}_{\text{nps}}$ is more complex: although originally designed to prevent the use of nonprintable colors in the physical world, in practice, the restricted color palette enforced by $\mathcal{L}_{\text{nps}}$ (defined in this article as a set of eight color) helps to maintain limited color diversity in camouflage. This reduces high-frequency texture details, thereby mitigating feature shifts under complex lighting conditions. Moreover, since high-frequency textures are difficult for cameras to capture in long-distance and dynamic scenarios, this color limitation, with its suppressive effect on high-frequency details, also facilitates better feature retention of camouflage across long-distance and dynamic settings.

To compare PBDR with the renderers used in other fully covered methods, we replaced the renderer in our approach with NMR (the renderer used in FCA [5] and DAS [16] method), which resulted in a significant drop in both CIR and DSR scores, as shown in Table V—(e). This is because PBDR integrates more real physical disturbances that can be actively introduced during training. Specifically, compared with NMR, PBDR's accurate simulation of global illumination and target reflectance characteristics effectively introduces physics-based texture disturbances during training, such as texture color shift disturbances (caused by strong illumination) and texture occlusion disturbances (caused by specular reflection), thereby guiding the camouflage to select texture features with greater physical robustness. Furthermore, PBDR also incorporates the simulation of other real physical effects, such as atmospheric scattering and motion blur, which further help to enhance the robustness of camouflage in long-distance and dynamic scenarios.

We also conducted an ablation study where we removed all of our proposed improvements and replaced the renderer with NMR. The results, shown in Table V—(g), indicate that the performance after testing was nearly identical to that of FCA. This further confirms that the proposed PBDR and PCTF can effectively enhance the camouflage robustness against real UAV-based detection tasks under multiscale, complex illumination, and dynamic scenarios.

In addition, we restricted the camouflage coverage to be consistent with patch. Compared with the complete method, the CAP score dropped by 37.12%, MVR by 16.21%, MSR by 5.79%, CIR by 9.28%, and DSR by 49.34% after testing. However, all scores remained higher than the camouflage generated by patch, with MSR, CIR, and DSR in particular showing improvements of 14.37%, 37.55%, and 210.31%, respectively. These results help disentangle the contributions of our proposed fully covered camouflage design from those of other optimization strategies. On the other hand, they also highlight the severe limitations of patch-based camouflage designs in coping with detection under dynamic scenarios.

As shown in Fig. 17, beyond the sport utility vehicle (SUV) model, we also transferred our camouflage to different types of vehicles (such as coupes and vans) and conducted experiments. The results show that the camouflaged coupe achieved an ASR of 86.46% against YOLOv5-OBB, while the camouflaged

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TABLE V

ABLATION EXPERIMENTAL RESULTS. THE CHOSEN DETECTION MODEL IS YOLOV5OBB. THE LEFT SIDE OF THE TABLE LISTS THE USAGE OF DIFFERENT LOSS FUNCTIONS OR FUNCTIONAL MODULES, WHILE THE RIGHT SIDE PRESENTS THE CORRESPONDING CALCULATED INDICATORS. THE ASR AND AP UNITS ARE EXPRESSED AS PERCENTAGES

	Proposed Losses			Proposed Modules			Evaluation Indicators				
	$\mathcal{L}_{ad}$	$\mathcal{L}_{nps}$	$\mathcal{L}_{dis}$	PBDR	F2T	Coverage	CAP	MVR	MSR	CIR	DSR
(a)	✓	✓	✓	✓	✓	Full	**89.26**	**90.11**	**89.66**	**94.00**	**84.49**
(b)	✓		✓	✓	✓	Full	80.46	86.62	82.68	82.48	72.61
(c)	✓	✓		✓	✓	Full	83.11	86.18	79.05	91.79	79.23
(d)	✓			✓	✓	Full	74.89	84.91	77.58	79.21	71.14
(e)	✓	✓	✓	NMR	✓	Full	71.62	85.86	80.37	64.68	60.83
(f)	✓	✓	✓	✓		Full	79.30	80.16	84.85	87.44	58.19
(g)	✓			NMR		Full	<u>50.33</u>	79.76	<u>72.06</u>	<u>62.79</u>	45.41
(h)	✓	✓	✓	✓	✓	Patch	56.12	<u>75.50</u>	84.47	85.28	<u>42.80</u>

van achieved an ASR of 79.84%. These findings further demonstrate the generalizability of our proposed method.

F. Discussion

This article identifies several technical limitations inherent in existing physical adversarial attack methods for remote sensing applications. To analyze and address these challenges, we first discuss the advantages and drawbacks of different types of physical adversarial camouflage. It is shown that most patch-based camouflages, which directly adopt the traditional digital adversarial attack framework, lack explicit modeling of the physical characteristics of both targets and operational conditions. Moreover, their patch design inevitably leaves camouflage blind zones. In contrast, fully covered camouflages exhibit greater design advantages in realistic remote sensing detection tasks characterized by varying viewing angles and temporal observations. To this end, we introduce a fully covered camouflage inverse rendering platform based on PBDR. Experimental results demonstrate that incorporating physics-driven disturbance factors, such as viewpoint changes, strong illumination, and reflection, into camouflage training enhances the physical robustness of the generated camouflage, thereby facilitating its transferability from laboratory-level evaluation to the complex environments of UAV operations. Finally, we analyze that the high-frequency texture details produced by conventional training strategies tend to be weakened or even lost at different imaging distances, leading to the degradation of adversarial effectiveness under the scale-varying and dynamic UAV detection scenarios. To overcome this issue, we propose the PCTF framework, which employs a hierarchical training strategy. The joint integration of PCTF and PBDR significantly improves the overall effectiveness of the generated camouflage.

V. CONCLUSION

This article investigates the problem of generating physical adversarial camouflage against UAV-based object detection models. Two novel approaches are introduced: PBDR and PCTF. PBDR enhances robustness under complex physical disturbances by proactively incorporating various simulated physical effects into the camouflage generation. PCTF leverages hierarchical training across data from different distances, combining interlayer training across scales and intralayer training across viewing angles to improve the multiscale and dynamic scenario adaptability of the generated camouflage. Extensive digital and physical experiments demonstrate that the proposed methods significantly increase ASR while maintaining robustness under varying illumination, viewpoints, distances, and dynamic scenarios. These findings highlight the importance of integrating physically driven interference factors and feature-preserving mechanisms when designing physical adversarial camouflage, providing new insights for developing robust attacks against UAV-based object detection systems.

As for limitations, the proposed framework does not yet include dedicated modules to improve cross-model transferability of camouflage. As a result, while the method shows strong adversarial performance in white-box settings, its effectiveness drops in black-box experiments. Future work will therefore aim to extend adversarial camouflage to broader task scenarios, including different intelligent vision tasks, target models, and camouflage objects, to enhance its generality.

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Ziyang Liu received the B.S. and M.S. degrees from China University of Geosciences, Beijing, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree in optical engineering with Harbin Institute of Technology, Harbin, China.

His research interests include computer vision and adversarial machine learning.

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Hongyuan Wang received the B.S. degree in thermal power engineering and the Ph.D. degree in optical engineering from Harbin Institute of Technology, Harbin, China, in 2000 and 2008, respectively.

He is currently a Professor with Harbin Institute of Technology. His major research interests are the application of deep learning methods in aerospace, photoelectric imaging system modeling and simulation, guidance, and navigation of spacecraft based on computer vision.



Qianhao Ning received the B.S. degree in photoelectric information science and engineering and the Ph.D. degree in optical engineering from Harbin Institute of Technology, Harbin, China, in 2019 and 2025, respectively.

He is currently a Post-Doctoral Fellow with Harbin Institute of Technology. His research interests include photoelectric imaging technology and photoelectric imaging system modeling and simulation.



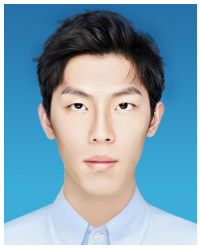
Zijian Wang received the B.S. degree in electrical engineering and automation from Shenyang University of Technology, Shenyang, China, in 2023. He is currently pursuing the Ph.D. degree in optical engineering with Harbin Institute of Technology, Harbin, China.

His research interests include computer vision and deep learning.



Yinxi Lu received the B.S. and M.S. degrees in optical engineering from Harbin Institute of Technology, Harbin, China, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision and 3-D reconstruction.



Yunzhao Zang received the B.S. degree in optical engineering from Harbin Institute of Technology, Harbin, China, in 2020, where he is currently pursuing the Ph.D. degree in optical engineering.

His research interests include all-time star sensor application, astronomical navigation, and image processing.



Zhiqiang Yan (Member, IEEE) received the B.S. and Ph.D. degrees in optical engineering from Harbin Institute of Technology, Harbin, China, in 2017 and 2023, respectively.

He is currently a Post-Doctoral Researcher in optical engineering with Harbin Institute of Technology. His research interests include imaging simulation, computer vision, and visual simultaneous localization and mapping (SLAM).

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